Geavanceerde computerarchitectuur: lab-1

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1 Learning objective

In this lab session we learn the basics of GPU programming with CUDA by completing a simple an array flip. By comparing the use of the GPU vs the use of the CPU we get to know the strengths of using the GPU.

2 Exercise

The code starts with an variable array size wich then gets filled with incremented numbers. This array alongside the empty flipped array gets passed to the flip function where the GPU gets chosen and memory in the GPU gets allocated for the two arrays. To make the process faster the GPU can split the work into blocks and perform these at the same time. The Threads have an id within their block and the block also had an id. Using this we can get each threads true index when combining the block id with the thread id and use this to flip this index in the array. Afterwards the result needs to be copied into the host memory from the GPU. Now we do the same but with the CPU to test the difference in speed.

3 Conclusions

When using large numbers the execute speeds between the two methods get larger. The GPU is the faster option between the two. The CPU is faster in the beginning because the GPU method needs to handle the memory allocation. The reason the GPU scores better is because the CPU only has a few cores so while it can also perform operations simultaneously it is limited to the small amount of cores it has compared to the GPU. The GPU is designed to excel at parallel execution so in this case it can assign more cores to the flipping of the array than the CPU which leads to a significantly quicker executing speed when using large enough numbers or in general operations. This is why a GPU is used in a computer, to offload some of the CPU workload better designed for the GPU. This means that the CPU is freed up to handle other tasks.